# What we're building — POC architecture

3 agents · 1 dashboard · 0 paid APIs

## Frontend

Vanilla HTML · CSS · JavaScript

- Alert queue dashboard
- Live case investigation view
- Manual name screening tool

## Backend — FastAPI + SQLite

Python · async · Server-Sent Events

- REST API: cases, alerts, screening
- Agent runner with real-time SSE streaming
- Audit log — every action timestamped

AI agents (run automatically when a case is triggered)

### AML agent

Transaction analysis

1. Pulls mock transaction history
2. Detects patterns (structuring, velocity, round-trips)
3. LLM scores risk + drafts SAR narrative

### Sanctions agent

OFAC SDN screening

1. Downloads OFAC SDN XML (free, updated daily)
2. Fuzzy name match via rapidfuzz in SQLite
3. LLM resolves true vs false positives

### PEP + EDD agent

OpenSanctions + Wikidata

1. OpenSanctions free API (PEP, crime, watchlists)
2. Wikidata SPARQL for entity & political roles
3. Adverse media via GNews RSS scrape

Free data sources: OFAC SDN XML · OpenSanctions API · Wikidata SPARQL · GNews RSS · Synthetic mock transactions (SQLite)

# How it works — end to end

1. **Alert comes in**

   Analyst submits a name or a flagged transaction triggers automatically from the mock AML queue.

2. **Agent picks it up**

   The orchestrator routes to the right agent (AML, Sanctions, or PEP). Steps stream live to the UI via SSE.

3. **Data gathered**

   Agent queries OFAC local cache, OpenSanctions API, Wikidata, and news — no paid vendors needed.

4. **Claude reasons**

   The Claude API scores risk, resolves false positives, and drafts an investigation narrative in plain English.

5. **Human reviews**

   Analyst sees the full evidence package and approves, rejects, or edits. Every action is logged to audit table.

# Build order

8 milestones, always something running

### Phase 1 — Foundation

- ✓ SQLite schema + seed data
- ✓ FastAPI skeleton (alerts, cases CRUD)

### Phase 2 — Sanctions

- ✓ OFAC XML downloader + local index
- ✓ Fuzzy name match agent + SSE stream

### Phase 3 — PEP + EDD

- ✓ OpenSanctions API integration
- ✓ Wikidata lookup + news scrape

### Phase 4 — Intelligence

- ✓ AML pattern detection agent
- ✓ Claude API: scoring + narrative draft